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ORIGINAL RESEARCH

Disturbance history, neighborhood crowding and soil conditions jointly shape tree growth in temperate forests

Shuai Fang[1,2] · Jing Ren[1,3] · Marc William Cadotte[4,5] · Zuoqiang Yuan[6] · Zhanqing Hao[6] · Xugao Wang[1,2] · Fei Lin[1,2] · Claire Fortunel[3]

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Abstract
Understanding how different mechanisms act and interact in shaping communities and ecosystems is essential to better predict their future with global change. Disturbance legacy, abiotic conditions, and biotic interactions can simultaneously influence tree growth, but it remains unclear what are their relative contributions and whether they have additive or interactive effects. We examined the separate and joint effects of disturbance intensity, soil conditions, and neighborhood crowding on tree growth in 10 temperate forests in northeast China. We found that disturbance was the strongest driver of tree growth, followed by neighbors and soil. Specifically, trees grew slower with decreasing initial disturbance intensity, but with increasing neighborhood crowding, soil pH and soil total phosphorus. Interestingly, the decrease in tree growth with increasing soil pH and soil phosphorus was steeper with high initial disturbance intensity. Testing the role of species traits, we showed that fast-growing species exhibited greater maximum tree size, but lower wood density and specific leaf area. Species with lower wood density grew faster with increasing initial disturbance intensity, while species with higher specific leaf area suffered less from neighbors in areas with high initial disturbance intensity. Our study suggests that accounting for both individual and interactive effects of multiple drivers is crucial to better predict forest dynamics.

Keywords Tree growth · Functional trait · Disturbance · Neighborhood crowding · Soil

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✉ Fei Lin
linfei@iae.ac.cn

✉ Claire Fortunel
claire.fortunel@ird.fr

1 CAS Key Laboratory of Forest Ecology and Management, Institute of Applied Ecology, Chinese Academy of Sciences, Shenyang, China

2 Key Laboratory of Terrestrial Ecosystem Carbon Neutrality, Liaoning, China

3 AMAP (Botanique et Modélisation de l'Architecture des Plantes et des Végétations), CIRAD, CNRS, INRAE, IRD, Université de Montpellier, Montpellier, France

4 Department of Biological Sciences, University of Toronto-Scarborough, Toronto, ON M1C 1A4, Canada

5 Department of Ecology and Evolutionary Biology, University of Toronto, Toronto, ON M5S 3B2, Canada

6 Research Center for Ecological and Environmental Sciences, Northwestern Polytechnical University, Xi'an, China

Introduction

Forests play an important role in providing fundamental ecosystem services to human populations locally and globally, such as storing carbon and maintaining biodiversity (Brockerhoff et al. 2017; Felipe-Lucia 2021). However, increasing evidences indicate that forests in many parts of the world are undergoing rapid changes due to anthropogenic disturbance (Barlow et al. 2016; Ezcurra 2016; Seidl et al. 2017). These changes have altered the structure and dynamics of forests, thereby reducing their ability to provide ecosystem services (Bengtsson et al. 2000; Seidl et al. 2014; Hubau et al. 2020). Better understanding the relative contributions of anthropogenic disturbance and natural ecological drivers to forest dynamics via their impacts on tree species vital rates (e.g. growth) is an important prerequisite for improved conservation and sustainable management practices of forest ecosystems (Sousa 1984; Attiwill 1994; Lindenmayer et al. 2000).

Disturbance history can play an important role in determining forest dynamics (Veblen et al. 1994). For instance, selective logging typically removes large individuals of

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